

AI-Powered Customer Support and Service Automation

Introduction

This project develops an AI chatbot to automate customer support, answer FAQs, track orders, and escalate complex issues.

Ideation

Problem: Slow support and high costs. Solution: AI virtual assistant available 24/7.

Requirement Analysis

Users, admin, chatbot, knowledge base, NLP engine, database.

Project Design

Architecture: User -> Chatbot -> NLP -> Database -> Response.

Planning

Requirement gathering, design, development, testing, deployment.

Testing

Functional, performance, and usability testing.

Results

Reduced response time, improved customer satisfaction, continuous availability.

Advantages

24/7 support, faster responses, cost reduction, scalability.

Disadvantages

Needs training data, limited handling of rare cases.

Conclusion

AI improves support efficiency while reducing workload.

Future Scope

Voice assistants, multilingual support, sentiment analysis.

Appendix

GitHub: N/A, Dataset: Customer support FAQ dataset.